

Bank Customer Churn Analysis

Data Analytics & Business Intelligence Project

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Tools Used: **Power BI, Python, Pandas, Matplotlib, Seaborn**

Dataset Size: **10,000 Customer Records**

Business Problem

Customer churn is a critical issue for retail banks because acquiring new customers is significantly more expensive than retaining existing ones.

The objective of this project is to identify the key factors that cause customers to leave the bank and provide data-driven recommendations to improve customer retention.

Business Objectives

The analysis aims to answer the following business questions:

- What is the overall customer churn rate?
- Which customer segments have the highest churn risk?
- How do demographic factors (age, gender, geography) influence churn?
- How do financial attributes (balance, salary, credit score) impact churn?
- What behavioral indicators (products used, activity level) predict churn?

Executive Summary

This report presents an in-depth analysis of customer churn patterns for a retail bank using Power BI and Python.

The project aims to identify key factors influencing customer attrition and develop actionable insights to improve retention strategies. The dataset includes 10,000 customer records containing demographic, financial, and behavioral variables, analyzed to uncover trends across geography, gender, age, and activity levels.

Dataset Overview

The dataset contains **10,000 customer records** from a retail bank.

Key features include:

Feature	Description
CreditScore	Customer credit rating
Geography	Country (France, Germany, Spain)
Gender	Male / Female
Age	Customer age
Tenure	Years with the bank
Balance	Account balance
NumOfProducts	Number of bank products used
IsActiveMember	Customer activity status
EstimatedSalary	Customer salary
Exited	Target variable (Churn / Not Churn)

Data Cleaning & Preparation

The dataset was prepared using **Python (Pandas)** before analysis.

Steps performed:

- Checked for **missing values**
- Removed **duplicate records**
- Validated **data types**
- Verified **data distributions**

Result:

The dataset contained **no missing values**, allowing direct analysis.

Exploratory Data Analysis (Python)

Python was used to perform statistical analysis and visual exploration.

Key Analysis Performed

- Distribution of churn vs retained customers
- Age distribution for churned customers
- Salary comparison between churned and retained customers
- Correlation analysis between numeric variables
- Gender, geography, and activity-based churn patterns

Example Findings

- Customers aged **40-60** show **significantly higher churn rates**
- **Inactive members are far more likely to churn**
- Customers using **only one product** show the **highest churn probability**

Business Intelligence Dashboard (Power BI)

An interactive **Power BI dashboard** was created to monitor churn metrics.

Key KPIs

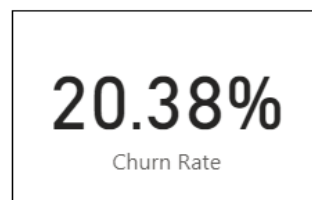
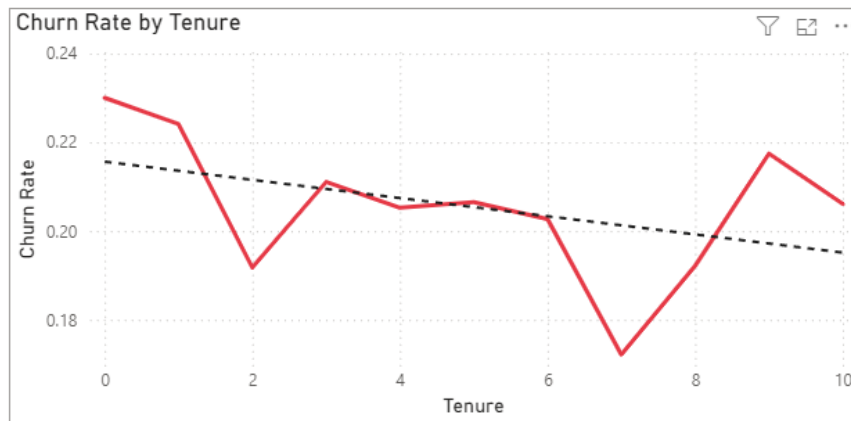
KPI	Value
Total Customers	10,000
Churned Customers	2,034
Churn Rate	20.34%
Active Customers	~51%

Country	Churn Rate
Germany	32.44%
France	16.15%
Spain	16.67%

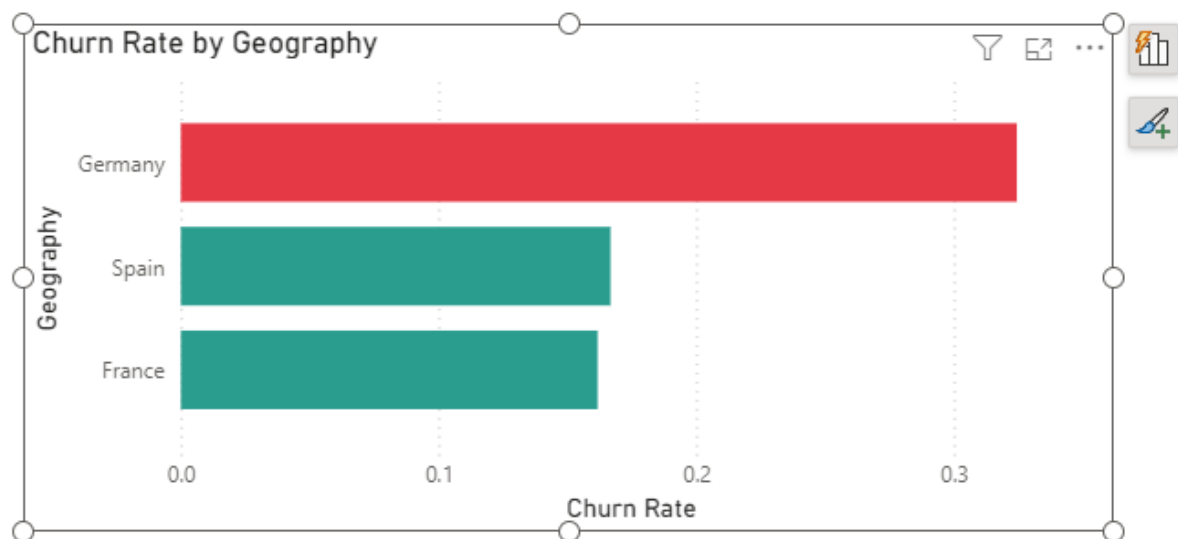
Germany shows **almost double the churn rate** compared to other regions

Section A: Power BI – Business Intelligence & Dashboards

1. What is the overall churn rate, and how does it trend over time (if applicable)?

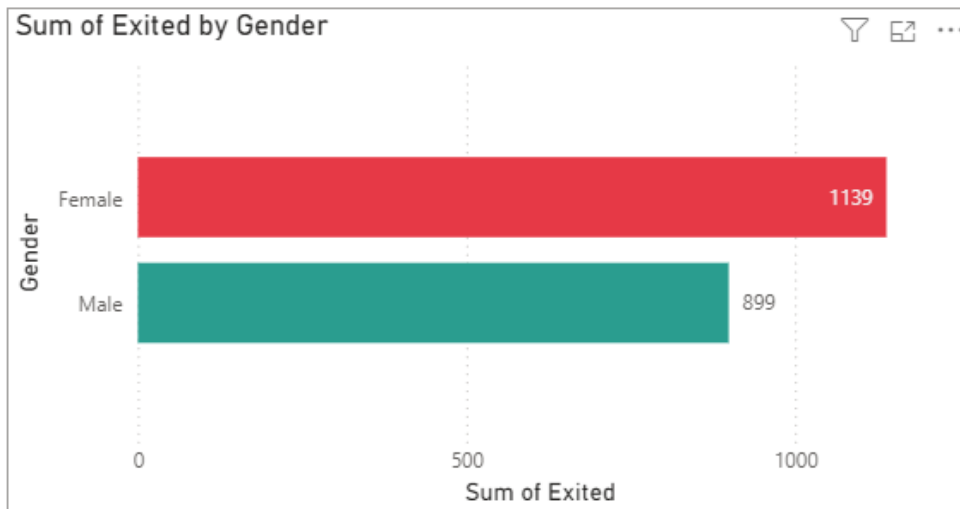


2. How does churn vary by geography (France, Germany, Spain)?



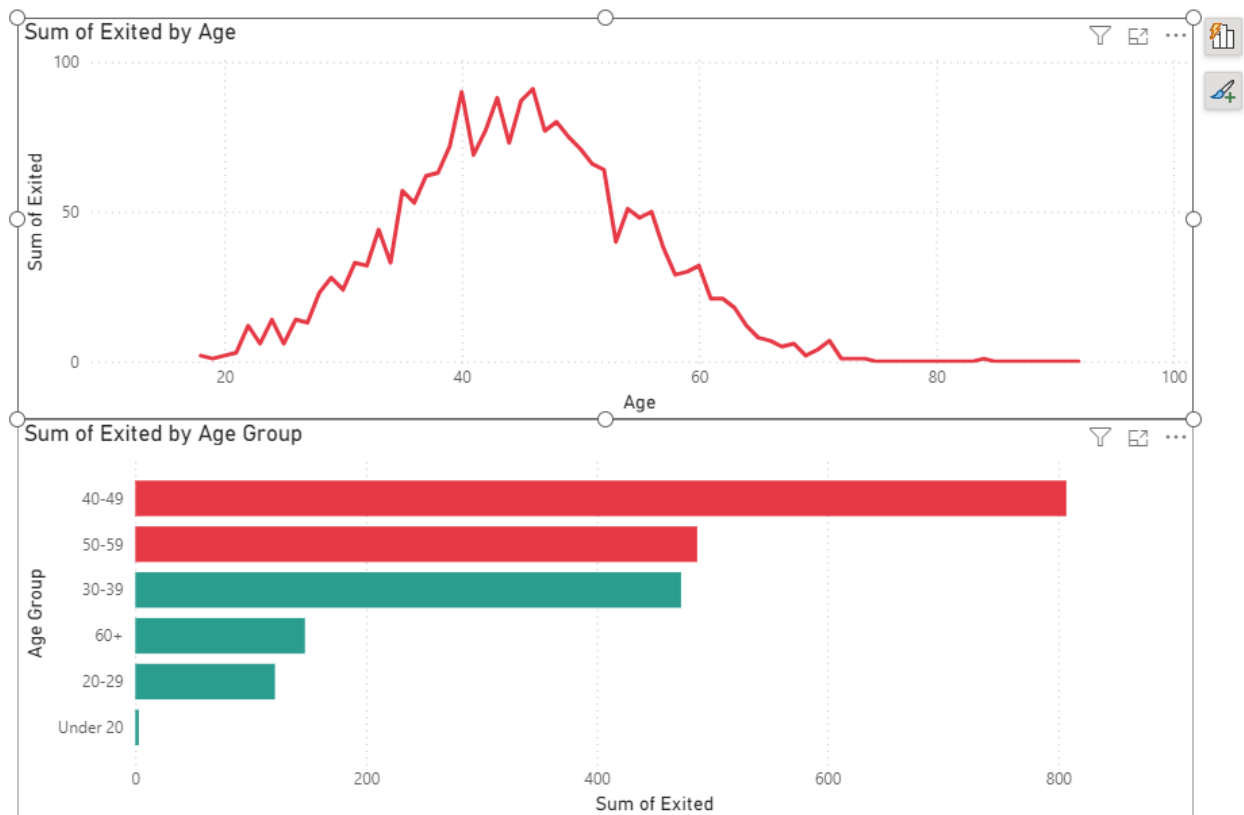
the churn rate by geography from your dataset: **France = 16.15%, Germany = 32.44%, Spain= 16.67%**

3. What are the differences in churn rates between male and female customers?



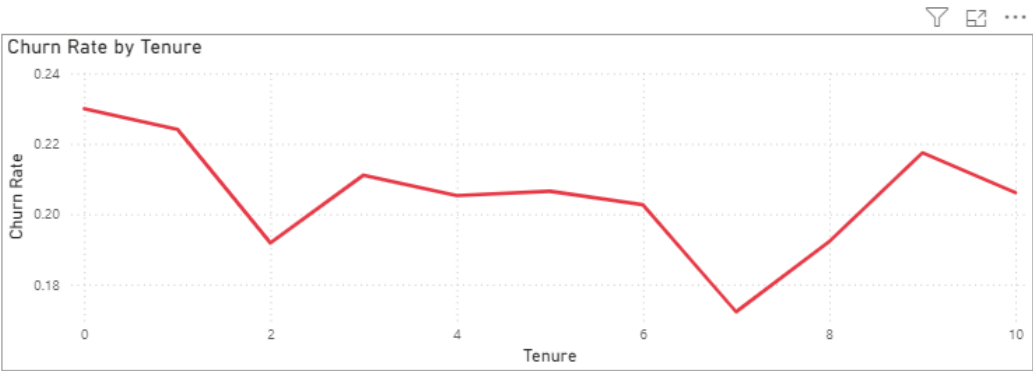
Churn Rate is Higher in Female (1139) then as compare to Male (899)

4. What is the churn rate distribution across different age groups?

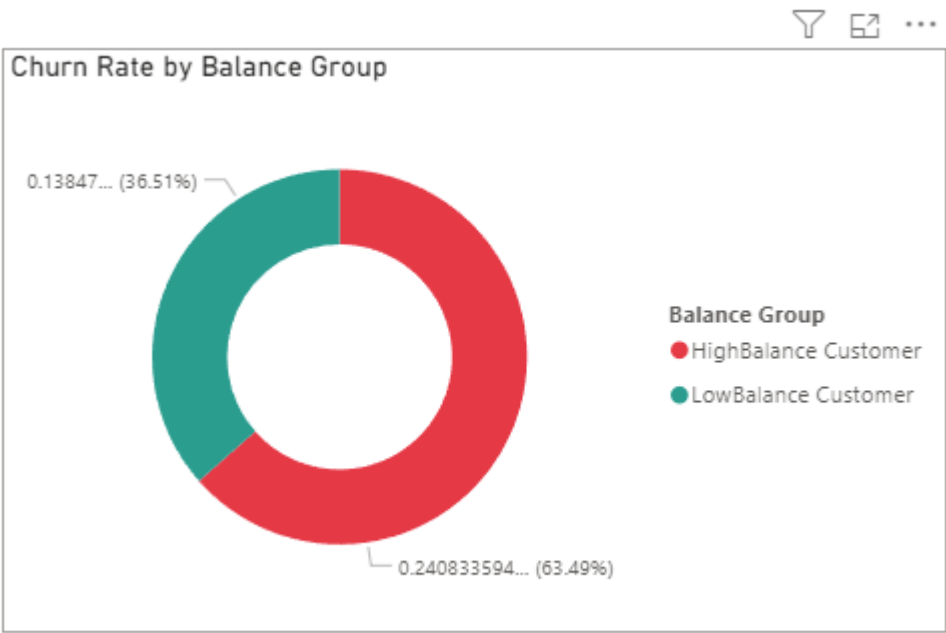


Age group from 40-49 has highest churn rate as per data

5. How does churn behavior change with tenure?

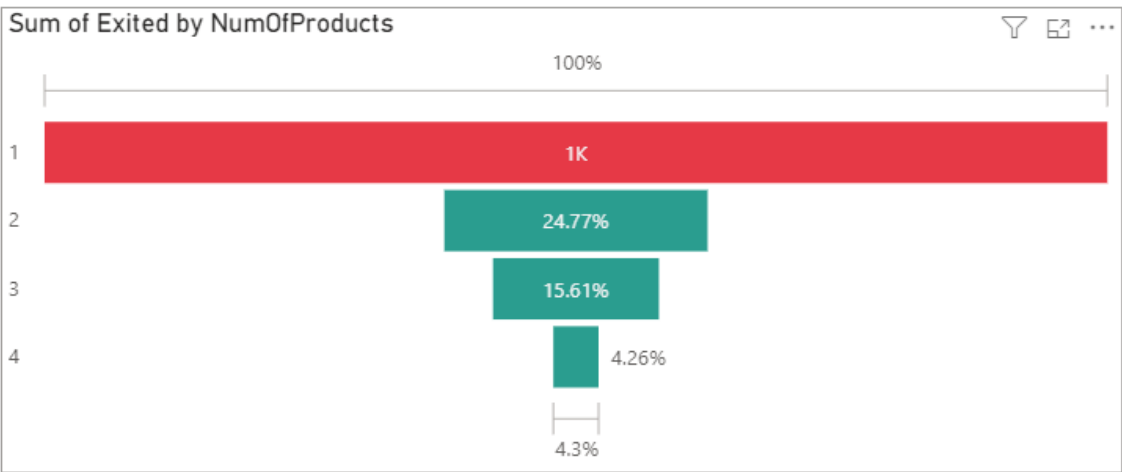


6. What percentage of high-balance customers (e.g., >\$100K) have churned?



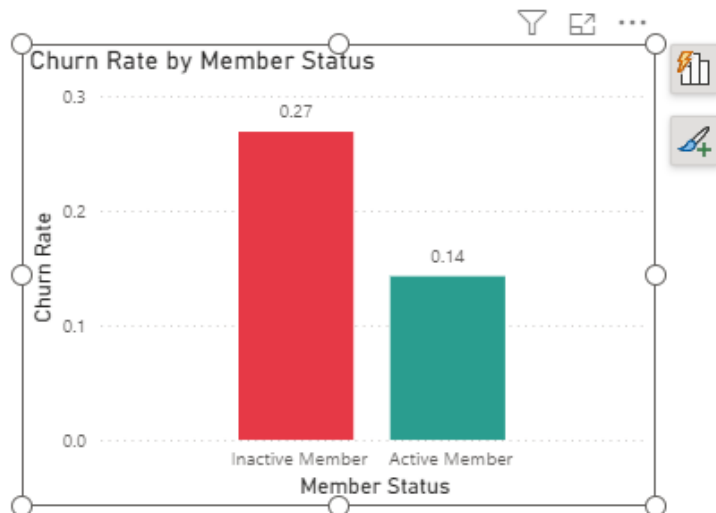
36.51% of High-balance Customers have churned.

7. How do churn rates compare across different NumOfProducts?

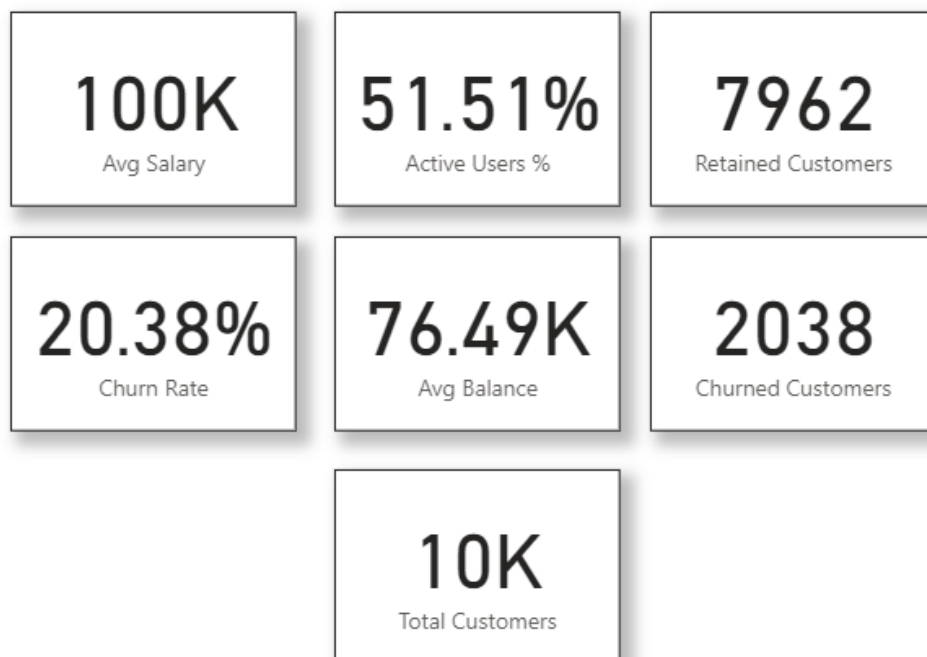


Customers using only one product show a significantly higher likelihood of churn.

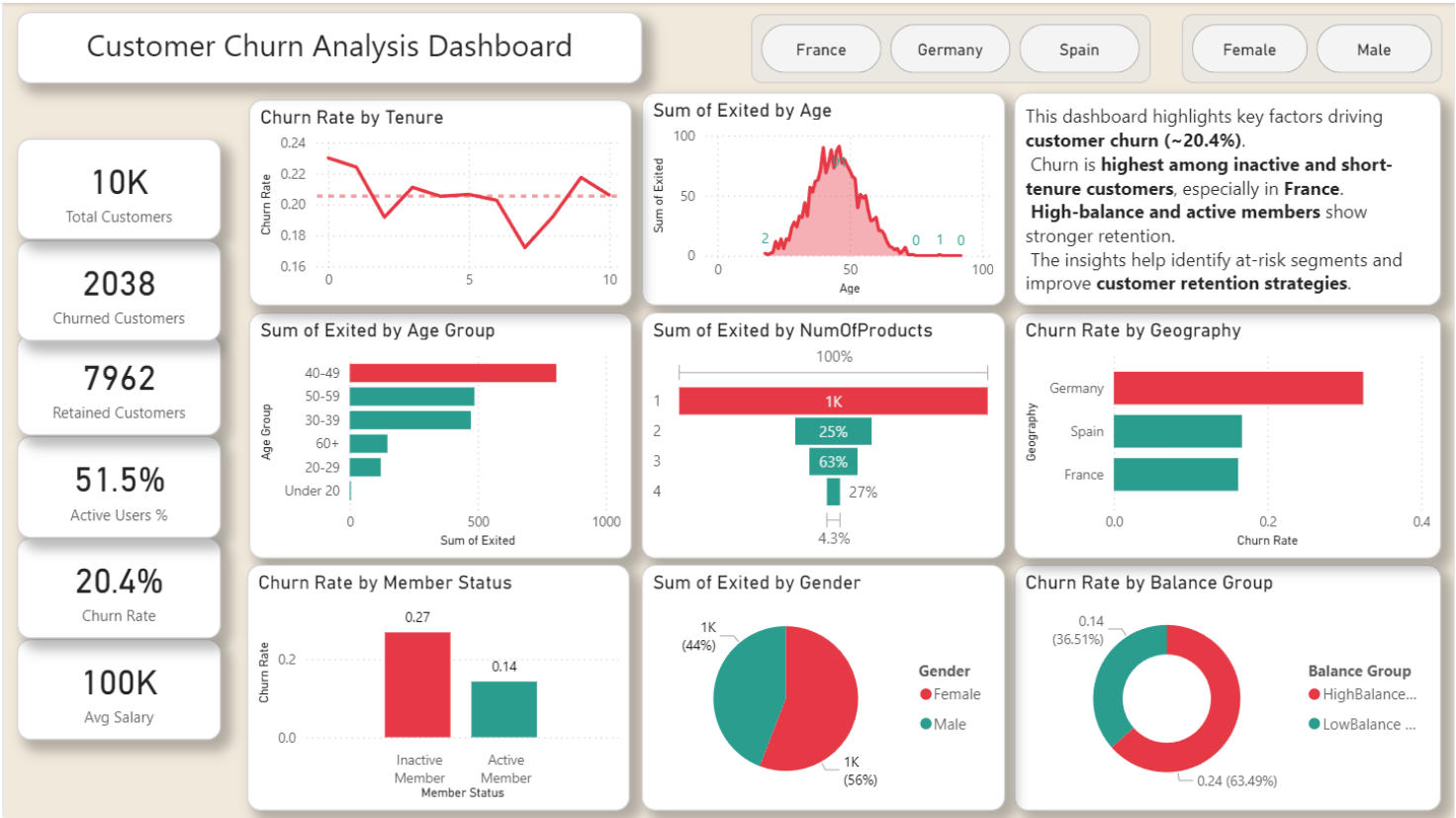
8. Are inactive members (IsActiveMember = 0) more likely to churn?



9. Can we build a KPI dashboard to track churn metrics like average salary, balance, and active users?



10. What does a customer persona dashboard (churned vs. retained) look like?



Section B: Python – Exploratory Data Analysis (Provide a description of your answers in all the cells explaining the outputs)

11. Check the percentage of missing data and handle accordingly.

```
d1.isna().sum()
```

...	0
RowNumber	0
CustomerId	0
Surname	0
CreditScore	0
Geography	0
Gender	0
Age	0
Tenure	0
Balance	0
NumOfProducts	0
HasCrCard	0
IsActiveMember	0
EstimatedSalary	0
Exited	0
Complain	0
Satisfaction Score	0
Card Type	0
Point Earned	0

dtype: int64

no missing data.

12. How many rows and columns are there in the dataset?

```
print("Rows:", df.shape[0])
print("Columns:", df.shape[1])
```

Rows: 10000
Columns: 18

13. What is the distribution of churned vs. non-churned customers?

```
[20] count_dis = df["Exited"].value_counts()
[21] count_dis.index = ["Non-Churned", "Churned"]
[22] percent_distribution = (count_dis / len(df)) * 100
print("Count Distribution:\n", count_dis)
print("\nPercentage Distribution:\n", percent_distribution.round(2))
```

... Count Distribution:
Non-Churned 7962
Churned 2038
Name: count, dtype: int64

Percentage Distribution:
Non-Churned 79.62
Churned 20.38
Name: count, dtype: float64

14. What is the distribution of EstimatedSalary of churned and retained Customers?

```
salary_distribution = df.groupby("Exited")["EstimatedSalary"].describe()
print(salary_distribution)
```

	count	mean	std	min	25%	50%	\
Exited							
0	7962.0	99726.853141	57399.956717	90.07	50783.295	99620.355	
1	2038.0	101509.908783	57932.623392	11.58	51924.020	102489.335	
		75%	max				
Exited							
0	148602.4450	199992.48					
1	152443.8575	199808.10					

15. How do churn rates vary by Gender, Geography, and IsActiveMember?

```
# Function to calculate churn rate
def churn_rate(group):
    return (group["Exited"].mean() * 100).round(2)

# Churn by Gender
churn_by_gender = df.groupby("Gender").apply(churn_rate)
print("Churn Rate by Gender:\n", churn_by_gender, "\n")

# Churn by Geography
churn_by_geo = df.groupby("Geography").apply(churn_rate)
print("Churn Rate by Geography:\n", churn_by_geo, "\n")

# Churn by Active/Inactive Members
churn_by_active = df.groupby("IsActiveMember").apply(churn_rate)
print("Churn Rate by Active/Inactive Members:\n", churn_by_active)
```

... Churn Rate by Gender:

Gender	
Female	25.07
Male	16.47

dtype: float64

Churn Rate by Geography:

Geography	
France	16.17
Germany	32.44
Spain	16.67

dtype: float64

Churn Rate by Active/Inactive Members:

IsActiveMember	
0	26.87
1	14.27

dtype: float64

16. What is the average CreditScore, Balance, and EstimatedSalary of churned vs. retained customers?

```
avg_values = df.groupby("Exited")[["CreditScore", "Balance", "EstimatedSalary"]].mean().round(2)

avg_values.index = ["Non-Churned", "Churned"]
```

avg_values

...

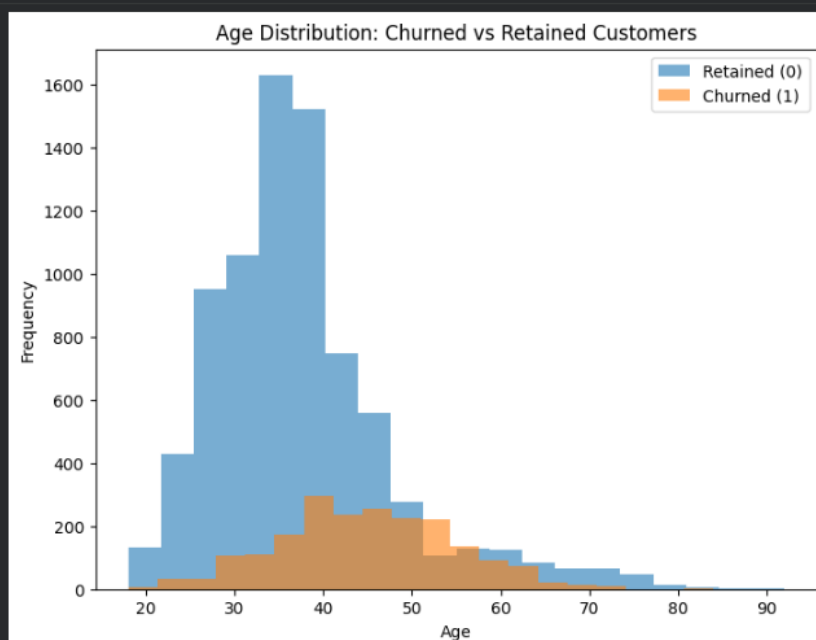
	CreditScore	Balance	EstimatedSalary
Non-Churned	651.84	72742.75	99726.85
Churned	645.41	91109.48	101509.91

Next steps: [Generate code with avg_values](#) [New interactive sheet](#)

17. How does Age impact churn? Plot histograms and boxplots for churned and non-churned groups.

```
Churned = df[df["Exited"] == 1] ["Age"]
Retained = df[df["Exited"] == 0] ["Age"]
```

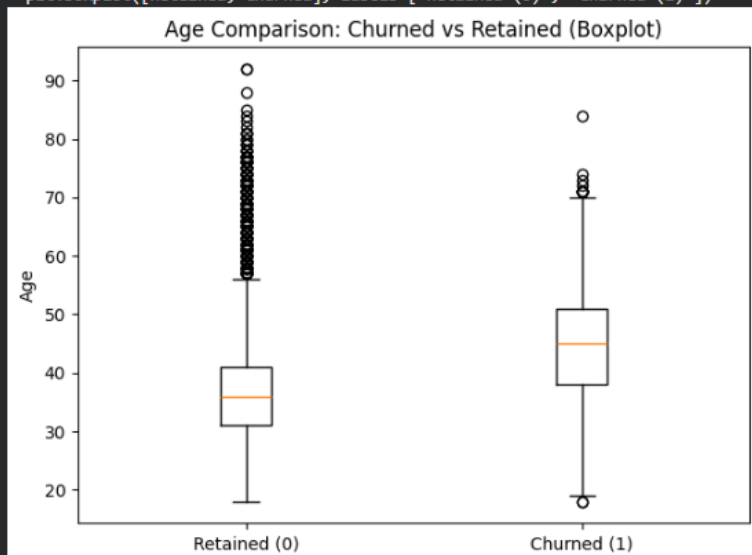
```
plt.figure(figsize=(8, 6))
plt.hist(Retained, bins=20, alpha=0.6, label="Retained (0)")
plt.hist(Churned, bins=20, alpha=0.6, label="Churned (1)")
plt.title("Age Distribution: Churned vs Retained Customers")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.legend()
plt.show()
```



```
plt.figure(figsize=(7, 5))
plt.boxplot([Retained, Churned], labels=["Retained (0)", "Churned (1)"])

plt.title("Age Comparison: Churned vs Retained (Boxplot)")
plt.ylabel("Age")
plt.show()
```

... /tmp/ipython-input-3827525609.py:2: MatplotlibDeprecationWarning: The 'labels' parameter
plt.boxplot([Retained, Churned], labels=["Retained (0)", "Churned (1)"])



18. Is there any correlation among numeric features like CreditScore, Balance, and EstimatedSalary?

```
[25] numeric_df = df[['CreditScore', 'Balance', 'EstimatedSalary']]
[26] corr_matrix = numeric_df.corr()
[27] corr_matrix
```

	CreditScore	Balance	EstimatedSalary
CreditScore	1.000000	0.006268	-0.001384
Balance	0.006268	1.000000	0.012797
EstimatedSalary	-0.001384	0.012797	1.000000

Next steps: [Generate code with corr_matrix](#) [New interactive sheet](#)

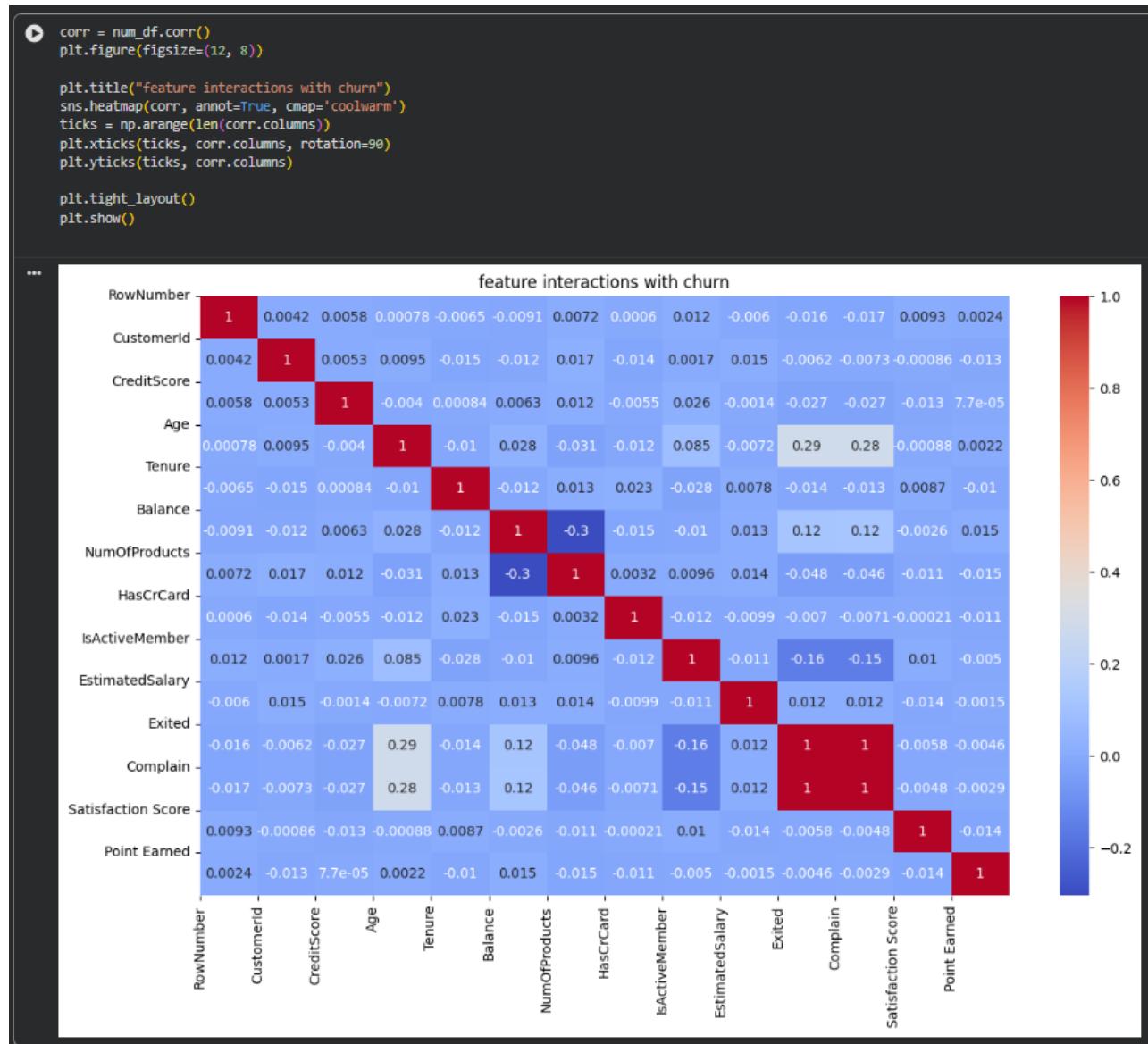
[25] Start coding or generate with AI.

19. Are customers with only one product (NumOfProducts = 1) more likely to churn than those with multiple?



- If **NumOfProducts = 1** has the highest churn rate → customers with only one product are more likely to churn.
- If **multiple-product customers** (2, 3, 4) have lower churn → they stay longer and are more loyal.

20.What does a heatmap reveal about feature interactions with churn?



Key Insights & KPI Highlights

- Overall churn rate: Approximately 20.34% of total customers.
- Geography: Germany showed the highest churn rate, followed by France and Spain.
- Gender: Female customers exhibited slightly higher churn rates compared to males.
- Age: Higher churn observed among customers aged 40-60.
- Tenure: Customers with shorter tenure (<5 years) were more likely to churn.
- High-Balance Customers: Despite higher balances, customers with >\$100K balance showed on table churn.
- Products: Customers with only one product had the highest churn rate.
- Inactive Members: Non-active customers (IsActiveMember = 0) were more likely to leave.

Customer Churn Persona

Using Power BI segmentation, the typical churn-risk customer profile is:

- Age: **40-55 years**
- Balance: **>\$100,000**
- Products: **1 product only**
- Activity Status: **Inactive**
- Location: **Germany**

These insights help the bank create retention-focused campaigns targeting this high-risk segment.

Business Recommendations

Based on the analysis, the bank should consider:

1. Cross-Selling Strategy

Encourage customers to adopt multiple banking products to increase retention.

2. High-Value Customer Retention

Introduce premium loyalty programs for customers with balances >\$100K.

3. Regional Strategy

Investigate service quality issues in Germany where churn is highest.

4. Customer Engagement

Develop digital engagement campaigns for inactive members.

5. Targeted Retention Campaigns

Focus retention programs on customers aged 40-55, the highest churn segment.

Project Impact

This analysis enables the bank to:

- Identify **high-risk churn segments**
- Improve **customer retention strategies**
- Increase **cross-selling opportunities**
- Reduce **revenue loss from high-value customers**